

# ARAV SINHA

Python Developer Intern Applicant | B.Tech Computer Science & Engineering

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## CAREER OBJECTIVE

B.Tech CSE student (6th semester) seeking a Python Developer internship. Proficient in backend fundamentals, API design, and automation, with practical experience engineering AI-assisted architecture using FastAPI, relational databases, caching systems, and containerized workflows.

## EDUCATION

### Amity University Patna

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Timeline: 2023 – Present | Current Semester: 6th | Expected Graduation: 2027

### Vidya Niketan School (CBSE), Patna

Senior Secondary (Class XII) — PCM

### Scholars Abode School (CBSE), Patna

Secondary (Class X)

## TECHNICAL SKILLS

- Core Languages:** Python, C, SQL, JavaScript
- Backend Frameworks:** FastAPI, RESTful Web Services
- Databases & Caching:** PostgreSQL, Redis, Relational Database Design (DBMS)
- Tools & DevOps:** Git, GitHub, Docker, Docker Compose, Nginx, VS Code
- Frontend Basics:** HTML5, CSS3, JavaScript, Tailwind CSS, React, Next.js
- Core Architecture:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Operating Systems, Computer Networks, Compiler Design, Theory of Computation

## TECHNICAL PROJECTS

### EdXplore — Academic Workspace & AI Assistant

Tech Stack: Python, FastAPI, PostgreSQL, Redis, Docker Compose, Nginx, Next.js, LangChain

Repository: [github.com/iaravsinha/student-management-ai](https://github.com/iaravsinha/student-management-ai)

- Architected a multi-service modern web application decoupling frontend, REST API backend, AI generation worker, and storage layers.
- Engineered production-ready FastAPI endpoints optimizing standard academic CRUD operations, workflow validation, and state tracking.
- Integrated robust LLM orchestration workflows via LangChain for processing context-aware student inquiries against system knowledge bases.
- Implemented high-performance persistent storage structures in PostgreSQL alongside an optimized Redis memory cache for session layer isolation.
- Dockerized modular service deployment environments utilizing isolated networks, environment injectors, and explicit service orchestrations.

## ADDITIONAL INFORMATION

- Target Alignment:** Python Developer Intern | Backend Engineer Intern | AI/ML Intern
- Core Capabilities:** Clean data modeling, efficient structural programming, asynchronous APIs, microservices configuration
- Languages:** English (Professional proficiency), Hindi (Native fluency)
- Availability:** Immediate availability for full-time or part-time Summer Internship opportunities